DHIKA MAHESA SYAPUTRA

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Jalan Makasar, Kecamatan Lubuk Begalung, Kota Padang

AI Engineer and Computer Science graduate who builds end-to-end LLM and computer-vision systems. Experienced in agentic RAG pipelines (LangGraph, LangChain, Langflow) with FAISS and ChromaDB vector search and structured-output prompt engineering, and in machine learning and deep learning models (XGBoost, YOLOv8, MobileNetV2) taken from training to deployment. Python-first, with REST API integration (Flask, FastAPI) and a research and data-analysis background, focused on building reliable, safety-aware AI products.

Work Experience

Department of Computer Science Laboratory, Universitas Riau - Pekanbaru Jun 2026 - Oct 2026

Research Assistant

- Built a YOLOv8 detector that screens oil palm plantation images for unhealthy (Ganoderma-affected) trees, reaching 99.5% recall and 99.4% mAP@0.5 on 230 test images (15,182 trees) and cutting inspection time from an estimated 2–5 minutes per tree (manual survey) to ~18 ms per image of ~66 trees.
- Trained a ResNet50 classifier to identify Ganoderma-infected oil palm trees, reaching 0.80 precision and recall on 656 annotated images.
- Co-developed an MLP Regressor model to predict free fatty acid (FFA) and Rendemen content in palm oil fruit from near-infrared (NIR) spectral data, achieving an R^2 of 0.78 and an RMSE of 6.03 on 64 samples from various capture angles.
- Co-conducted research and contributed in preparing scientific article on deep learning subject for oil palm plantations; manuscript status: in preparation.

Dinas Komunikasi, Informatika, dan Statistik Provinsi Sumatera Barat - Padang Jan 2025 - Mar 2025

Data Analyst

- Built 4+ interactive dashboards (Python, Tableau, Excel) giving provincial officials real-time visibility into agriculture, livestock, and plantation output and farmer welfare.
- Engineered a data pipeline using Pandas and Excel that cleaned, transformed, and merged datasets from more than 20 sources (BPS, internal portals), closing data-availability gaps and cutting preparation time by 50%.
- Redesigned the province's main dashboard with dynamic filters and PDF export, reducing report-sharing effort for stakeholders from 5 manual steps to a single click.
- Developed a custom regional performance index (Indeks Kinerja Daerah) and Tableau geospatial maps covering 19 regencies/cities to surface productivity and sector-distribution trends.

Education.

Universitas Riau - Pekanbaru, Riau Aug 2022 - Jun 2026

Bachelor of Computer Science, 3.88/4.00

- Awardee of Digital Talent Scholarship (DTS) 2024 – Ministry of Communication and Informatics (Kominfo).
- Astra Honda SDGs Future Leaders 2025 (national, 100+ participants): selected participant; designed SatuRute, a community ride-sharing prototype to cut carbon emissions and congestion in Pekanbaru, and analyzed traffic and demographic data with Python and Tableau to identify high-demand zones.

AI and Machine Learning Projects

CiteCheck AI - Agentic RAG Citation Verification Assistant - Maju Bareng AI by Hacktiv8: LLM-Based Tools and Gemini API Integration for Data Scientists

Python, LangGraph, LangChain, FAISS and ChromaDB, Llama 3.3 70B and GPT-OSS 120B, Gradio

- Built a LangGraph agent with conditional routing (Retrieve → Evaluate → External Search) that checks whether thesis claims are supported, contradicted, or unrelated to uploaded papers, returning corrected sentences backed by source evidence.
- Designed strict exact-match evaluation prompts to prevent over-helpful hallucination, and added an LLM query rewriter that turns unsupported claims into academic keywords and retrieves alternative papers in real time via the Semantic Scholar API.
- Implemented ChromaDB and FAISS retrieval with Google Generative AI embeddings for the corpus collection, plus conversation memory (MemorySaver) for follow-up questions; validated on 25 claim–paper pairs with 84% verdict accuracy.

Kontrak Aman – Employment Contract Risk Checker for Fresh Graduates - IBM Skillsbuild University Education Program: Legal - AI Agent for Legal

Langflow, Gemini 3.6 Flash, JSON parser and Structured Output

- Designed a five-stage LLM pipeline (file ingestion → structured clause extraction → parser → risk-analysis LLM → chat output) that

converts raw employment contracts into a 10-category report with a 1–10 risk score, plain-language red flags, and negotiation questions for HR.
- Validated on a purpose-built fictitious contract seeded with 6 red-flag clauses (uncompensated probation, unpaid overtime); each clause was checked against Indonesian labor law (UU Ketenagakerjaan), detecting 6/6 and rating the contract 8/10 (high risk).
- Gave fresh graduates without legal training or budget for counsel a free, minutes-long second opinion before signing; added responsible-AI safeguards: an explicit not-legal-advice disclaimer, outputs grounded in regulation text, and testing on synthetic contract data only.

Moon-Phase Classification & Fishing Recommendation System - **Undergraduate Thesis**

TensorFlow, MobileNetV2, TFLite, Android (Java, MVVM), SQLite, Room Database, Stormglass API

- Achieved 88.32% test accuracy on 8-class moon-phase classification by fine-tuning an ImageNet-pretrained MobileNetV2 through 3 progressive transfer-learning phases, evaluated on 291 held-out images.
- Integrated real-time ocean data from the Stormglass API into a rule-based engine with 5 recommendation levels.
- Reduced the model size from 22 MB to 5 MB (77%) via TensorFlow Lite conversion and deployed it in an offline-first Android app, running inference in 100-200 ms per image.
- Manuscript under review at the Indonesian Journal of Electrical Engineering and Informatics (IJEEI, Scopus-indexed, Q3).

Oil Palm Fruit Ripeness Classification

Python, XGBoost, scikit-learn, Pandas

- Classified oil palm fruit ripeness (unripe vs. ripe) from two physicochemical features, moisture content and beta-carotene, using XGBoost, achieving 86% accuracy and 0.86 macro F1 on a balanced set of 50 samples.
- Evaluated with Leave-One-Out Cross-Validation to get a near-unbiased performance estimate on a small dataset, where a fixed train/test split would leave too few samples to test reliably.

Training and Technical Skills

- **Hard Skills**: LLMs & Prompt Engineering (Gemini, OpenAI GPT-OSS, Llama 3.3 70B, Claude), RAG & Agentic AI (FAISS and ChromaDB vector database, embeddings, LangGraph, LangChain, Langflow), Machine Learning and Deep Learning (supervised and unsupervised learning algorithm, object detection), MLOps, RESTful APIs (Flask, FastAPI, Spring Boot), external API integration (Semantic Scholar, marine-data API), Android and web server deployment, Gradio and Streamlit, Google Colab, Oracle Cloud, Google Cloud Platform (GCP), Git/GitHub, Python, Java (OOP), SQL, Tableau, Looker Studio, Excel, Docker CI/CD,
- **Soft Skills**: Analytical Thinking, Problem Solving, Data- Driven Decision Making, Effective Communication, Team Collaboration, Adaptability.
- **Pelatihan** ✓ (2024): Oracle Cloud Infrastructure 2023 Certified Foundations Associate
- **Pelatihan** ✓ (2024): Oracle Cloud Infrastructure Foundation I & II - Talent Scouting Academy Digital Talent Scholarship
- **Pelatihan** ✓ (2025): Data Classification and Summarization in the Student Developer Initiative Program - Hacktiv8
- **Pelatihan** ✓ (2026): IBM Skillsbuild University Education Program: Legal - AI Agent for Legal - Hacktiv8
- **Pelatihan** ✓ (2026): Maju Bareng AI Program: LLM-Based Tools and Gemini API Integration for Data Scientists -Hacktiv8
- **Pelatihan** ✓ (2026): Belajar Dasar AI - Dicoding Academy
- **Pelatihan** ✓ (2025): Belajar Dasar Data Science - Dicoding Academy
- **Pelatihan** ✓ (2025): Memulai Pemrograman dengan Python - Dicoding Academy
- **Pelatihan** ✓ (2025): Belajar Dasar Structured Query Language (SQL) - Dicoding Academy
- **Pelatihan** ✓ (2023): Java Programming - Oracle Academy
- **Pelatihan** ✓ (2023): Java Fundamentals - Oracle Academy